

Advanced (Habanero)**Q1.** What is the expansion of $(x + y)^2$?

- (A) $x^2 + 2xy + y^2$
- (B) $x^2 - 2xy + y^2$
- (C) $x^2 + xy + y^2$
- (D) $x^2 - xy + y^2$
- (E) $x^2 + y^2$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
- (B) $x^2 - 4x - 4$
- (C) $x^2 + 4x - 4$
- (D) $x^2 - 4x + 4$
- (E) $x^2 + 4$

Q3. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
- (B) $(x - 3)^2$
- (C) $(x + 3)(x - 3)$
- (D) $(x + 2)(x + 4)$
- (E) $(x - 2)(x - 4)$

Q4. What is the expansion of $(x - y)^2$?

- (A) $x^2 + 2xy + y^2$
- (B) $x^2 - 2xy + y^2$
- (C) $x^2 + xy + y^2$
- (D) $x^2 - xy + y^2$
- (E) $x^2 + y^2$

Q5. Which of the following is the factored form of $x^2 - 8x + 16$?

- (A) $(x + 4)^2$
- (B) $(x - 4)^2$
- (C) $(x + 2)(x + 6)$
- (D) $(x - 2)(x - 6)$
- (E) $(x + 2)(x - 6)$

Q6. What is the expansion of $(2x + 3y)^2$?

- (A) $4x^2 + 12xy + 9y^2$
- (B) $4x^2 - 12xy + 9y^2$
- (C) $4x^2 + 6xy + 9y^2$
- (D) $4x^2 - 6xy + 9y^2$
- (E) $4x^2 + 9y^2$

Q7. Which of the following is a perfect square trinomial?

- (A) $x^2 + 10x + 25$
- (B) $x^2 - 10x - 25$
- (C) $x^2 + 10x - 25$
- (D) $x^2 - 10x + 25$
- (E) $x^2 + 25$

Q8. What is the factored form of $x^2 + 12x + 36$?

- (A) $(x + 6)^2$
- (B) $(x - 6)^2$
- (C) $(x + 3)(x + 9)$
- (D) $(x - 3)(x - 9)$
- (E) $(x + 3)(x - 9)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 8x + 16$ and show all steps.**Q10.** Expand the expression $(3x - 2y)^2$ and simplify.**Chef's Tips**

- Read each question carefully.
- Use the space provided for each question.
- Show all work and explain your reasoning.